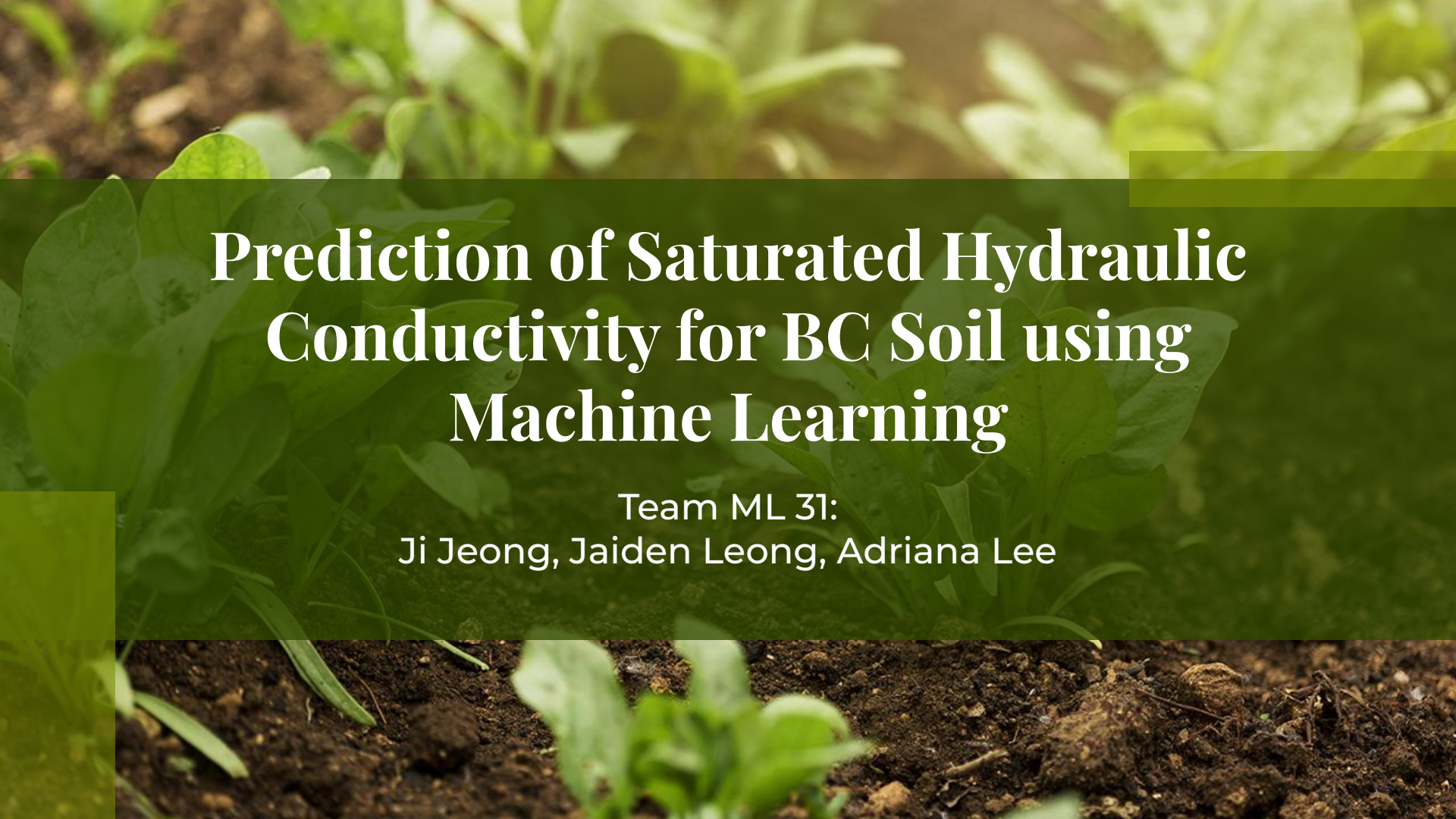
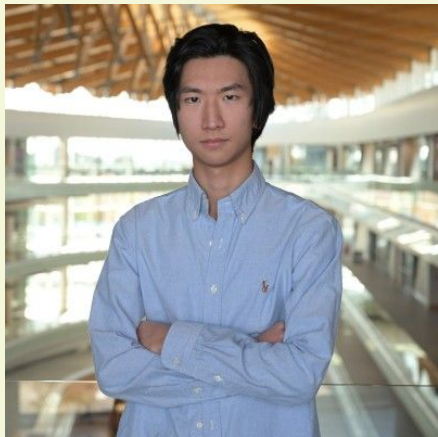




# Prediction of Saturated Hydraulic Conductivity for BC Soil using Machine Learning

Team ML 31:  
Ji Jeong, Jaiden Leong, Adriana Lee

# Team ML 31



Ji Jeong



Jaiden Leong



Adriana Lee

**01** **Problem Description**  
What is Saturated Hydraulic Conductivity ( $K_{Sat}$ )?

**02** **Importances**  
Why predict  $K_{Sat}$ ?

**03** **Solutions**  
How to predict  $K_{Sat}$ ?

**04** **Objectives**  
General and Machine Learning Objectives

**05** **Methods**  
How will it be solved?

**06** **Conclusions**  
Results of our solution and future improvements.

# What is $K_{Sat}$ ?

Saturated Hydraulic Conductivity ( $K_{Sat}$ ):

- Describes how easily water can move through the soil.
- Influences applications such as
  - Groundwater flow
  - Water infiltration and drainage
  - Soil Moisture



# Why Predict $K_{Sat}$ ?

Current Methods for determining  $K_{Sat}$ :

- Uses lab equipment and field instruments:
  - Permeameter
  - Tensiometer
  - CT Scanning<sup>[5]</sup>
- Estimates the  $K_{Sat}$  value:
  - Darcy's Law
  - Pedotransfer Function (PTF)

Allows for specific water targeting watering (Precision Agriculture):

- Saves water
- Saves farmers money
- Improves crop yield

# How to Predict $K_{Sat}$ ?

- Machine Learning (ML):
  - Supervised Learning
  - Unsupervised Learning
  - Semi-supervised Learning
  - Reinforcement Learning
  - Transfer Learning
  - Ensemble Learning



# How to Predict $K_{Sat}$ ?

- Machine Learning (ML):
  - Supervised Learning
    - Regression
    - Neural Network
  - Unsupervised Learning
  - Semi-supervised Learning
  - Reinforcement Learning
  - Transfer Learning
  - Ensemble Learning

# Objectives

## General

- Use BC Soil Data.
- Find accessible and affordable methods to obtain  $K_{Sat}$  values.
- Prevent water wastage.
- Improve crop yield.

## Machine Learning

- Accuracy
- Computation Time



# Methods

# Data Cleaning

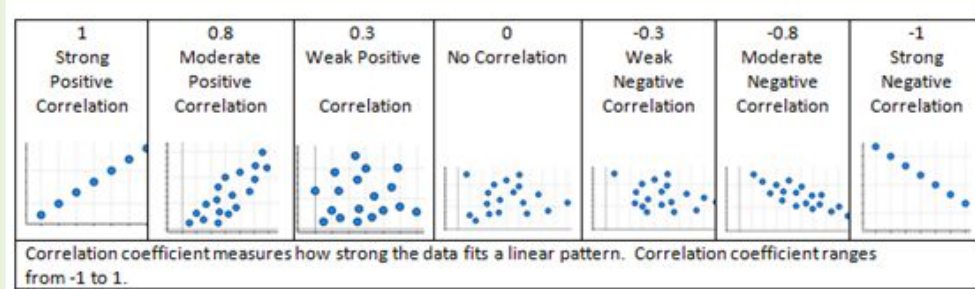
- Soil Sample data from Government of BC
- Reviewed for best correlation with  $K_{Sat}$ :
  - Silt, sand, clay, and coarse fragment contents
  - Water retention at varying pressures
- Parameters deemed essential in previous research:
  - Silt, sand, clay, coarse fragment, organic content
  - Bulk density
  - Electric conductivity
- These features are much easier to measure on site.



# Feature Selections

Dataset: “British Columbia Detailed Soil Survey - Soil Layer File”

- **Raw Features:** All 43 Features.
- **Correlation Features:** Top 7 features based on the highest correlations with  $K_{Sat}$ .
- **Literature Features:** Prominent features from previous research.
- **Optimal (Common) Features:** Features common in both correlation and literature features.



## Models

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Support Vector Machine for Regression (SVR)
- K-Nearest Neighbours (KNN)
- Boosted Regression Tree (BRT)

## Performance Metrics

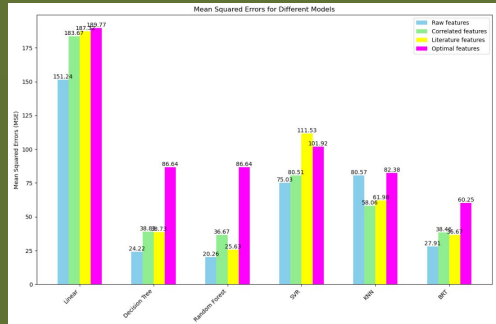
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Square Error (RMSE)
- Coefficient of Determination ( $R^2$ )
- Training Time
- 10-fold Cross Validation

# Results

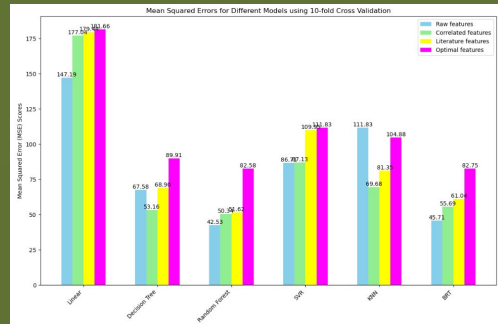
Random Forest Regressor model using the Raw data features showed the most stable performance.

## Performance Metric Summary

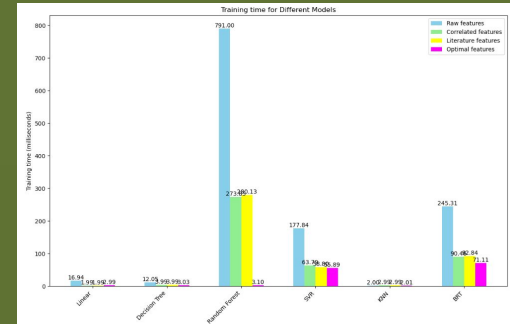
Mean Square Error for Different Models and Feature Sets



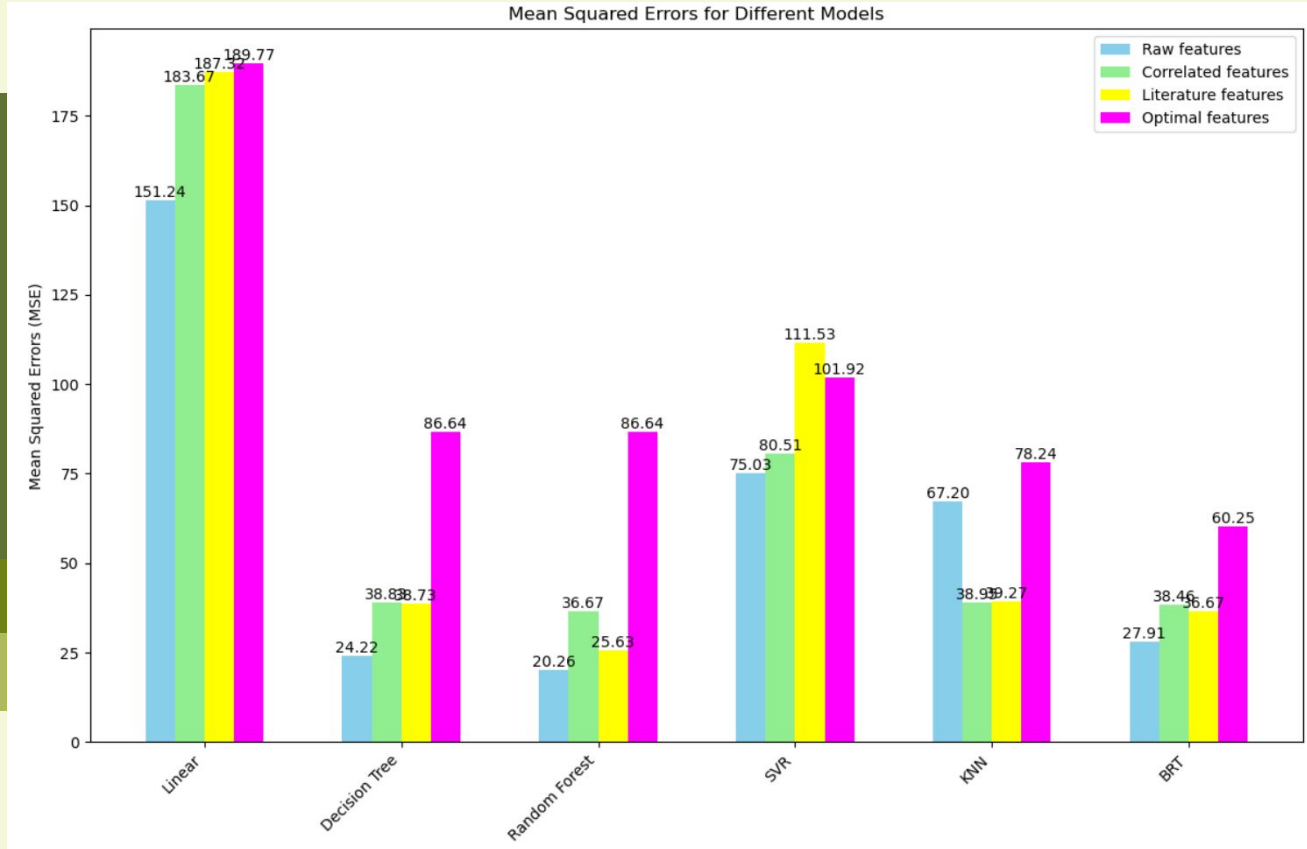
Mean Square Error for Different Models Using 10 fold Cross-Validation



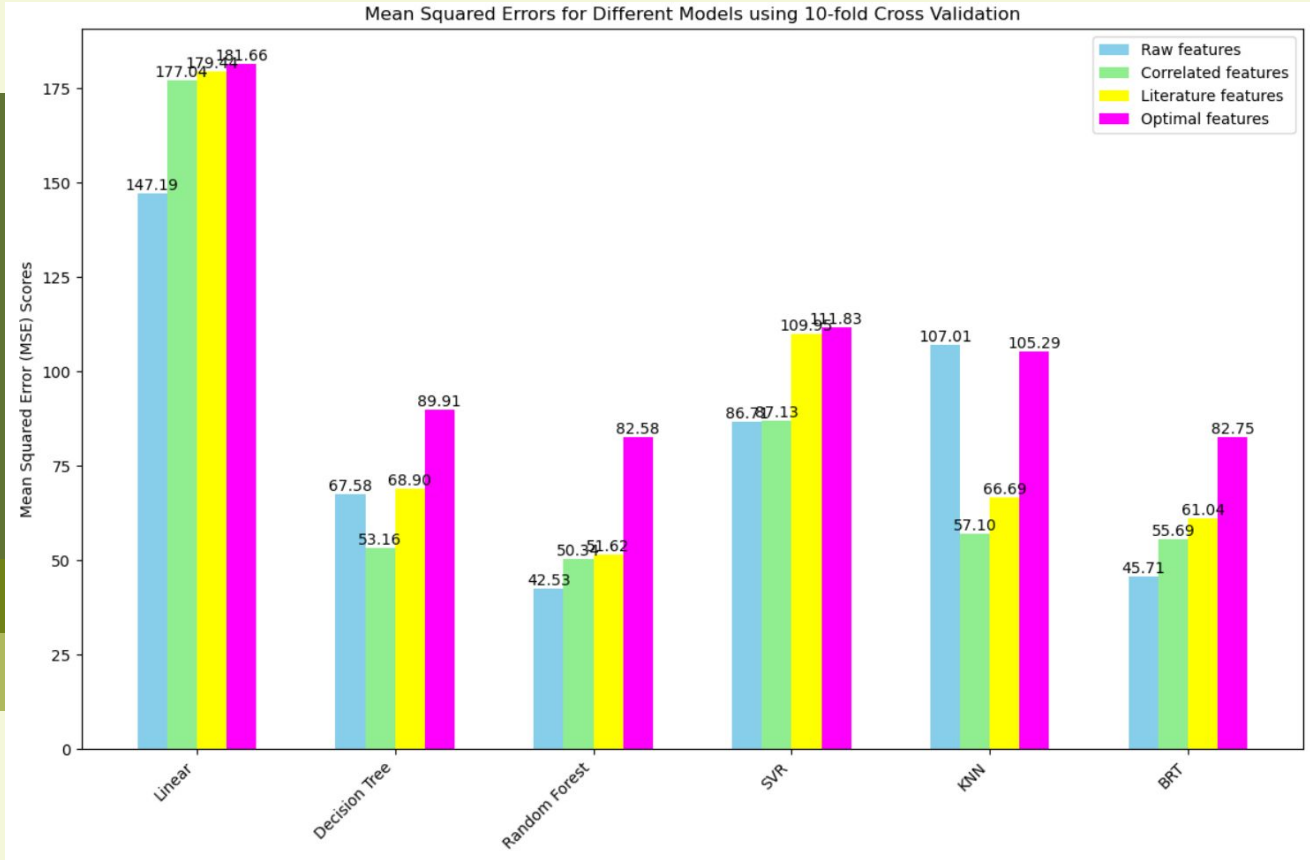
Training Time Across Different Models and Feature Sets



# Mean Square Errors for Different Models and Feature Sets

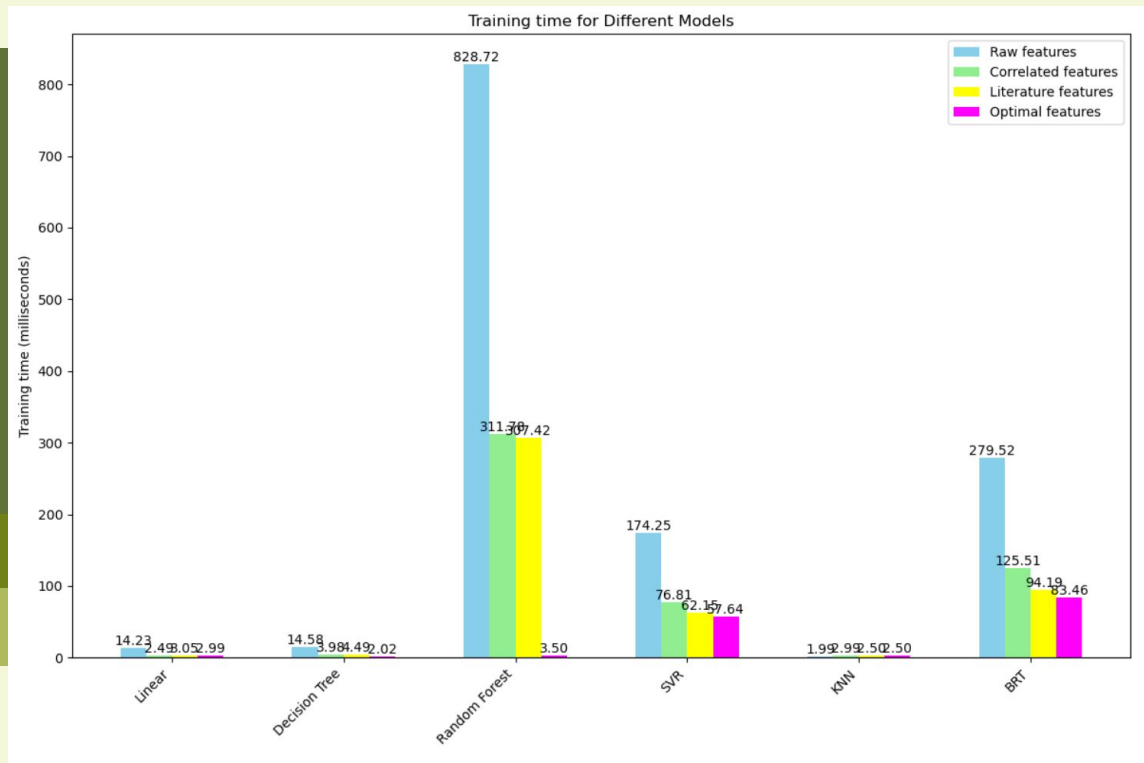


# Mean Square Error Scores for Different Models Using 10-fold Cross Validation





# Training Time Across Different Models and Feature Sets



# Conclusion

After modelling **6 different ML models** each trained and tested with **4 different feature sets**, our optimal model was found: **Random Forest Regressor** with raw data features such as:

- COFRAG
- TSAND
- TSILT
- TCLAY

# Conclusion

## Future improvements:

Include climate data from the Government of BC

- Would require the use of Geography APIs in order to convert the map units indicated in the soil samples to locations more easily comprehensible

Using the climate data, examine how weather patterns impact other soil properties such as:

- pH levels
- Soil erosion

# Conclusion

## Future improvements:

Consideration of Multicollinearity

- Adds weight to properties that are similar or dependent
- Inflates standard errors (e.g., MSE) and coefficient of determination ( $R^2$ )

In future models:

- Conduct correlation analysis between the parameters
- Feature Engineering
  - Principal Component Analysis
  - Feature Addition

# Conclusion

Using this project, farmers can:

- Easily determine  $K_{SAT}$  for specific areas of their plots using cheaper equipment on-site
  - Sieves
  - Visual inspection
- Allows for accessible precision agriculture at any level of farming in or outside BC

# Thank You

*Any questions?*

# References

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# Prediction of saturated hydraulic conductivity for BC soil using machine learning

Ji Jeong, Jaiden Leong, Adriana Lee

School of Mechatronic Systems Engineering, Simon Fraser University

## Introduction

Saturated hydraulic conductivity ( $K_{Sat}$ ) is a critical factor in agriculture, indicating how well soil can transmit water. Traditional methods for measuring  $K_{Sat}$  often rely on costly and inaccessible equipment or assumptions. By developing a machine learning (ML) algorithm that predicts  $K_{Sat}$  using real British Columbia (BC) soil data, local farmers can efficiently allocate water resources with greater accessibility and accuracy.

## Materials and Methods

Six ML algorithms were evaluated to identify the most effective predictor for  $K_{Sat}$  based on soil sample data collected across British Columbia, Canada. These algorithms include: Linear Regression, Decision Tree Regressor, Random Forest Regressor, Support Vector Regressor (SVR), K-Nearest Neighbours (KNN), and Boosted Regression Tree (BRT). The models were trained using four different sets of features derived from the BC soil sample data:

- All 43 Features
- Top seven features with highest correlation to  $K_{Sat}$
- Features commonly found in previous research studies
- Features shared between the top seven correlated features and academic literature

## Results

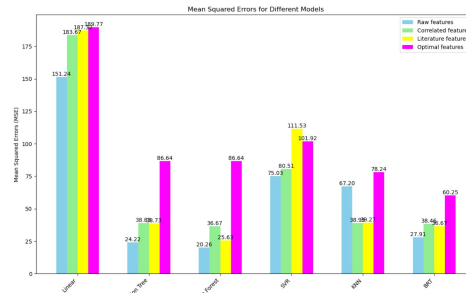


Figure 1: Bar charts of Mean Squared Errors (MSE) for different models and features sets.

Figure 2 displays the MSE scores obtained after conducting 10-fold cross-validation to assess the models' performance on new, unseen data. Interestingly, the results do not entirely align with those in Figure 1. Here, higher MSE values indicate better performance due to the negative MSE values returned from the cross-validation scores using the scikit-learn library in Python. Surprisingly, the linear regression model with overlapping features exhibits the highest performance, while the RF model shows the lowest performance with all 43 features. Consequently, a comprehensive analysis comparing the metrics from both Figure 1 and Figure 2 is necessary to determine the optimal model.

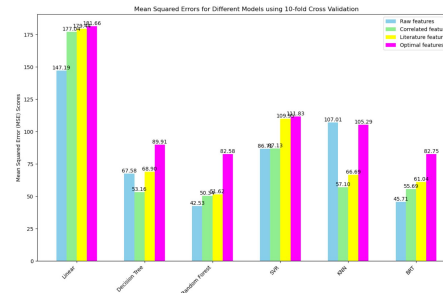


Figure 2: Bar charts of Mean Squared Errors (MSE) for different models and features sets after 10-fold cross validation.

## Conclusions

Based on the findings from Figures 1 and 2, it becomes evident that the Random Forest Regressor Model with the raw dataset features, demonstrated optimal performance. With minimal values in MSE, RMSE, and MAE and a  $R^2$  very close to 1, this model emerges as a strong contender for predicting  $K_{Sat}$  in BC soil. For future improvements, leveraging climate data sourced from the Government of BC to assess its influence on soil properties and to refine predictive models for  $K_{Sat}$ , holds promise for advancing the realm of accessible precision farming.

## Acknowledgements

We extend our sincere thanks to Dr. Mohammad Narimani from the School of Mechatronic Systems Engineering at Simon Fraser University for inspiring our exploration into creating a soil predictor using the soil data from BC, Canada, with the aim of supporting local farmers. Additionally, we express our gratitude to the Government of BC for providing access to the soil data online, enabling our research endeavors.

## Further information

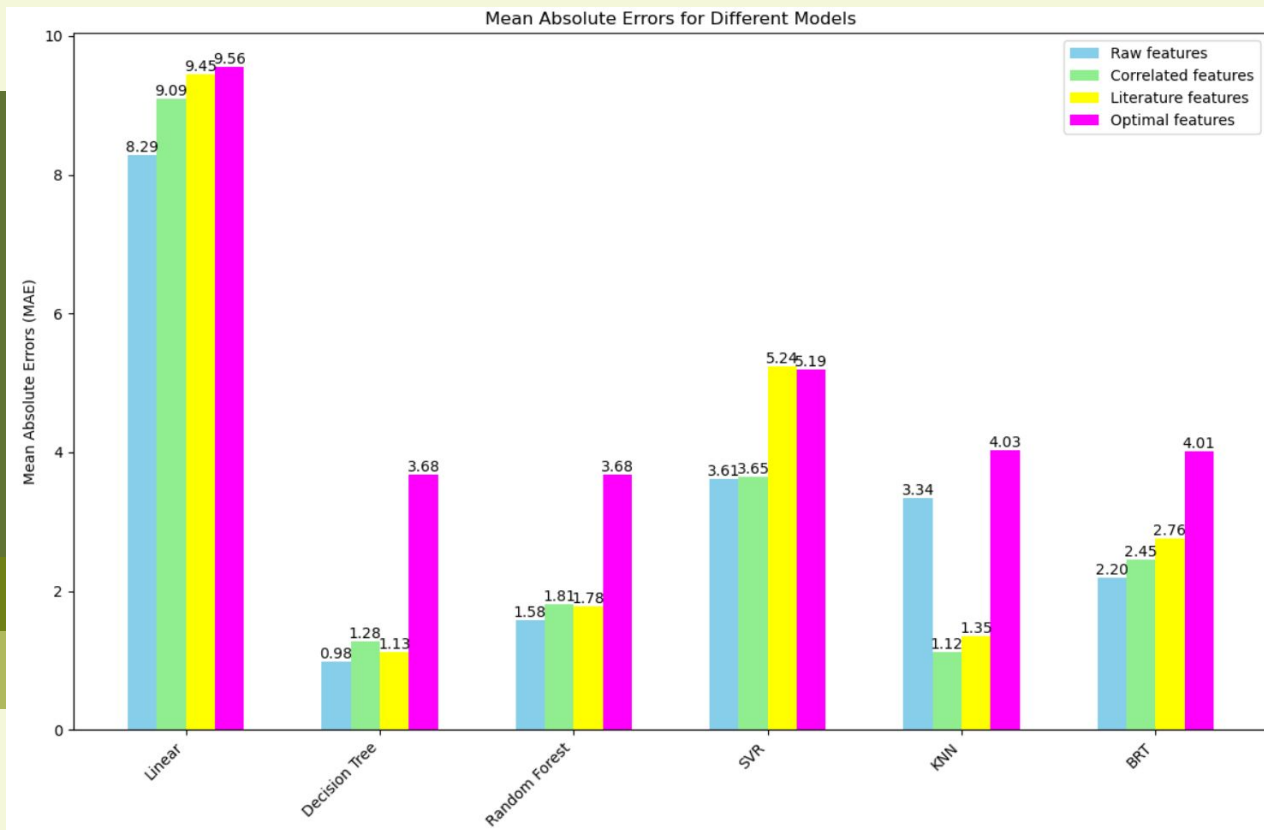
To learn more about this research, please contact:

- Ji Jeong: [ji\\_jeong@sfu.ca](mailto:ji_jeong@sfu.ca)
- Jaiden Leong: [jl152@sfu.ca](mailto:jl152@sfu.ca)
- Adriana Lee: [ala236@sfu.ca](mailto:ala236@sfu.ca)

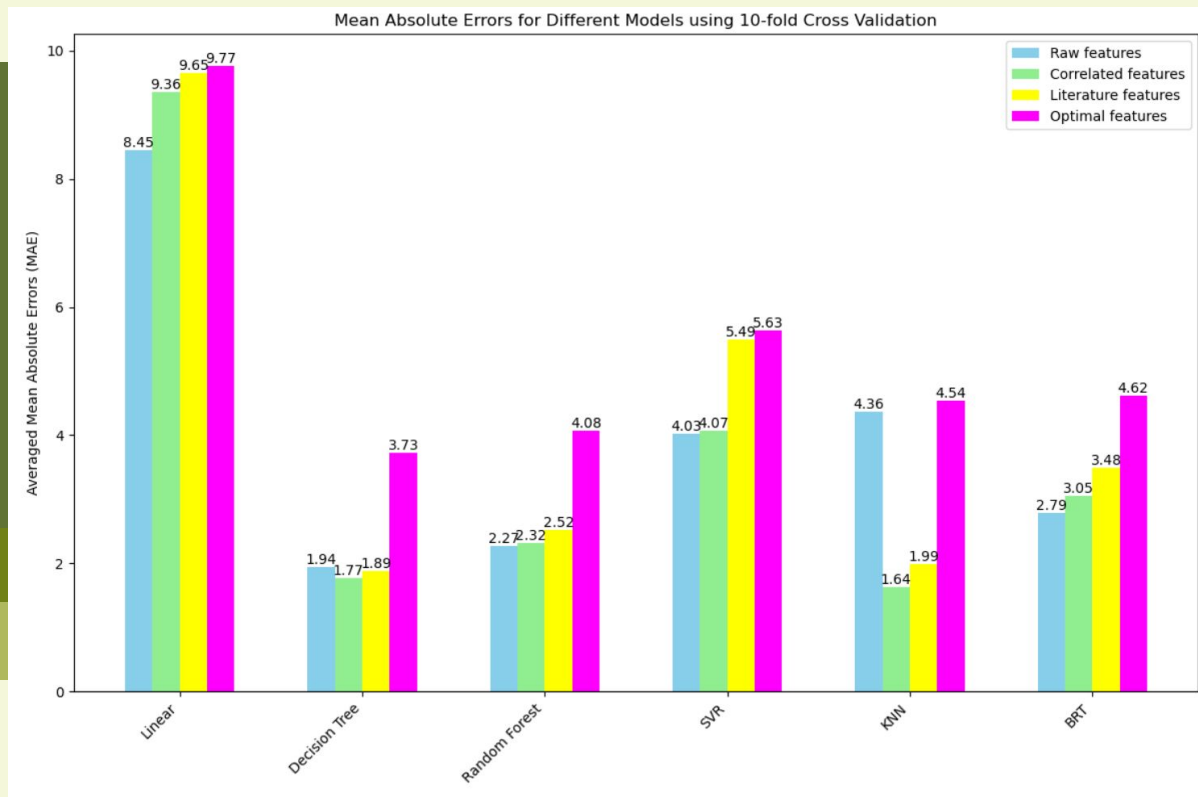


# Appendix

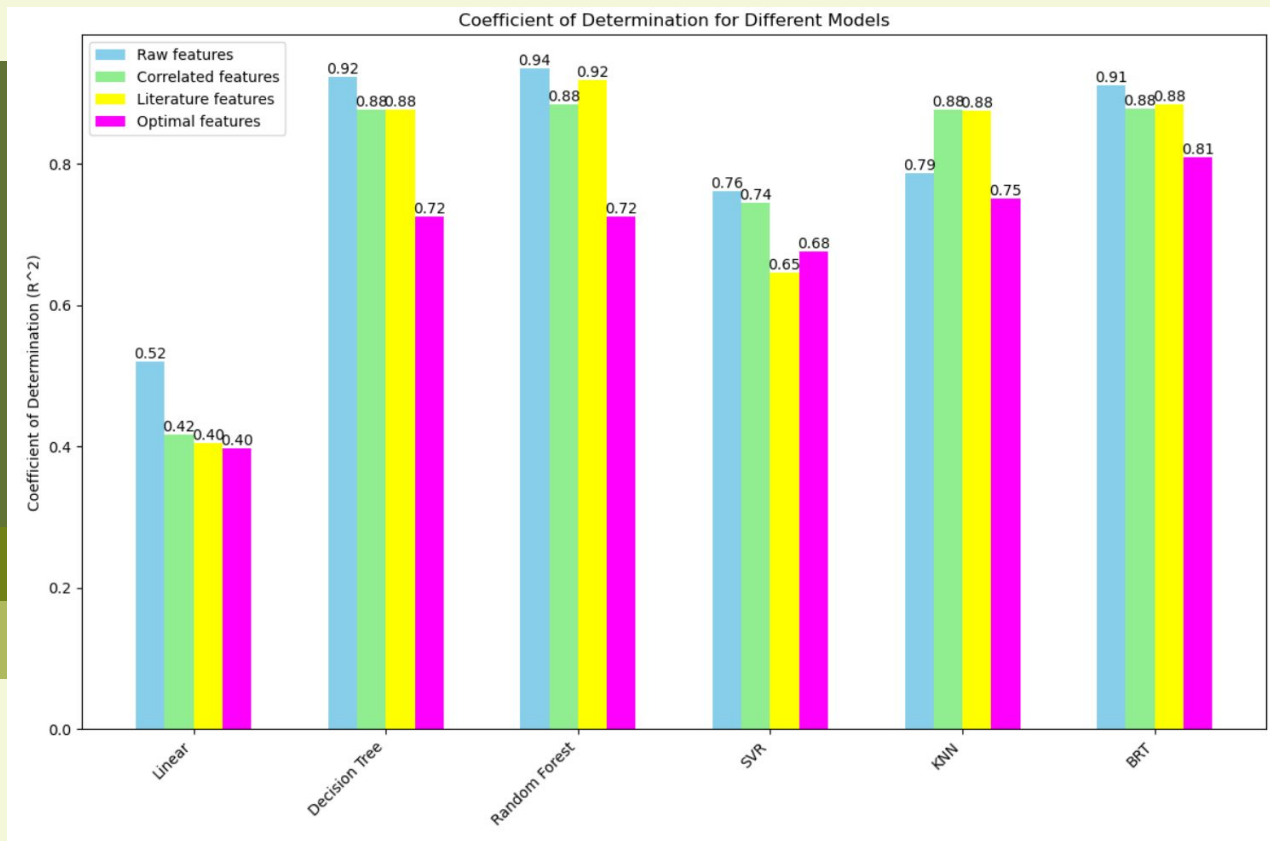
# Mean Absolute Errors for Different Models and Feature Sets



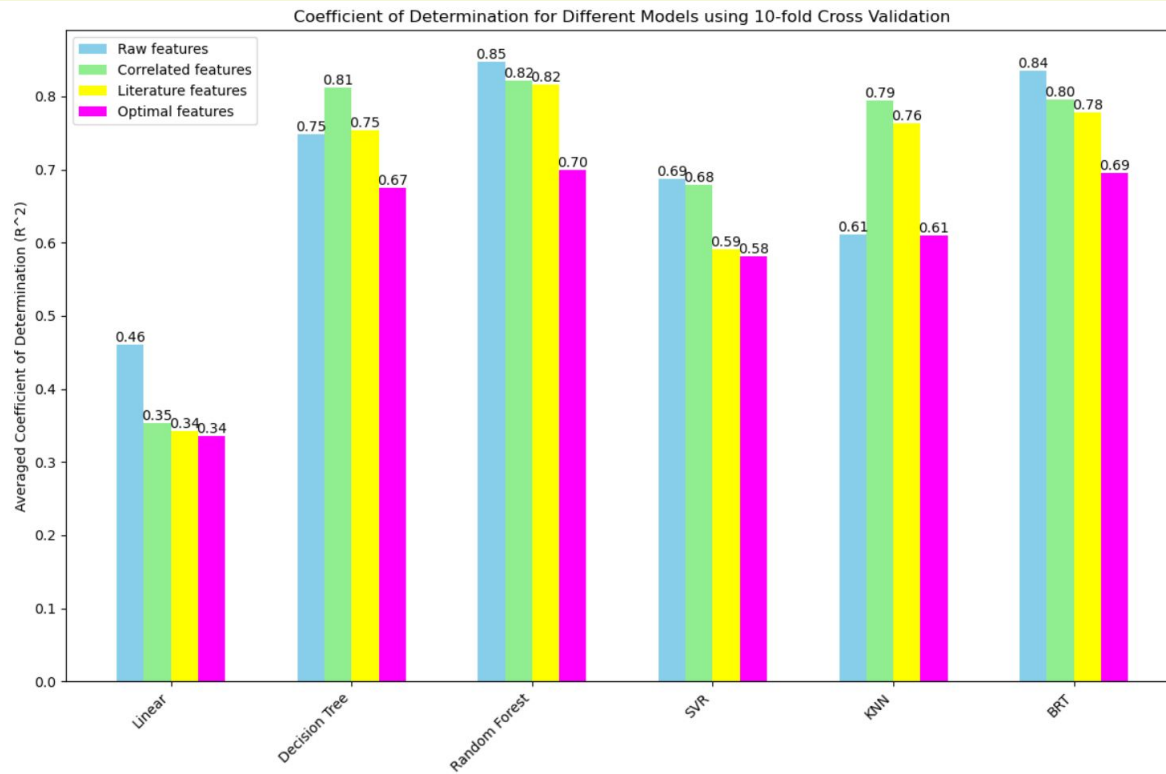
# Mean Absolute Errors for Different Models and Feature Sets Using 10-fold Cross Validation



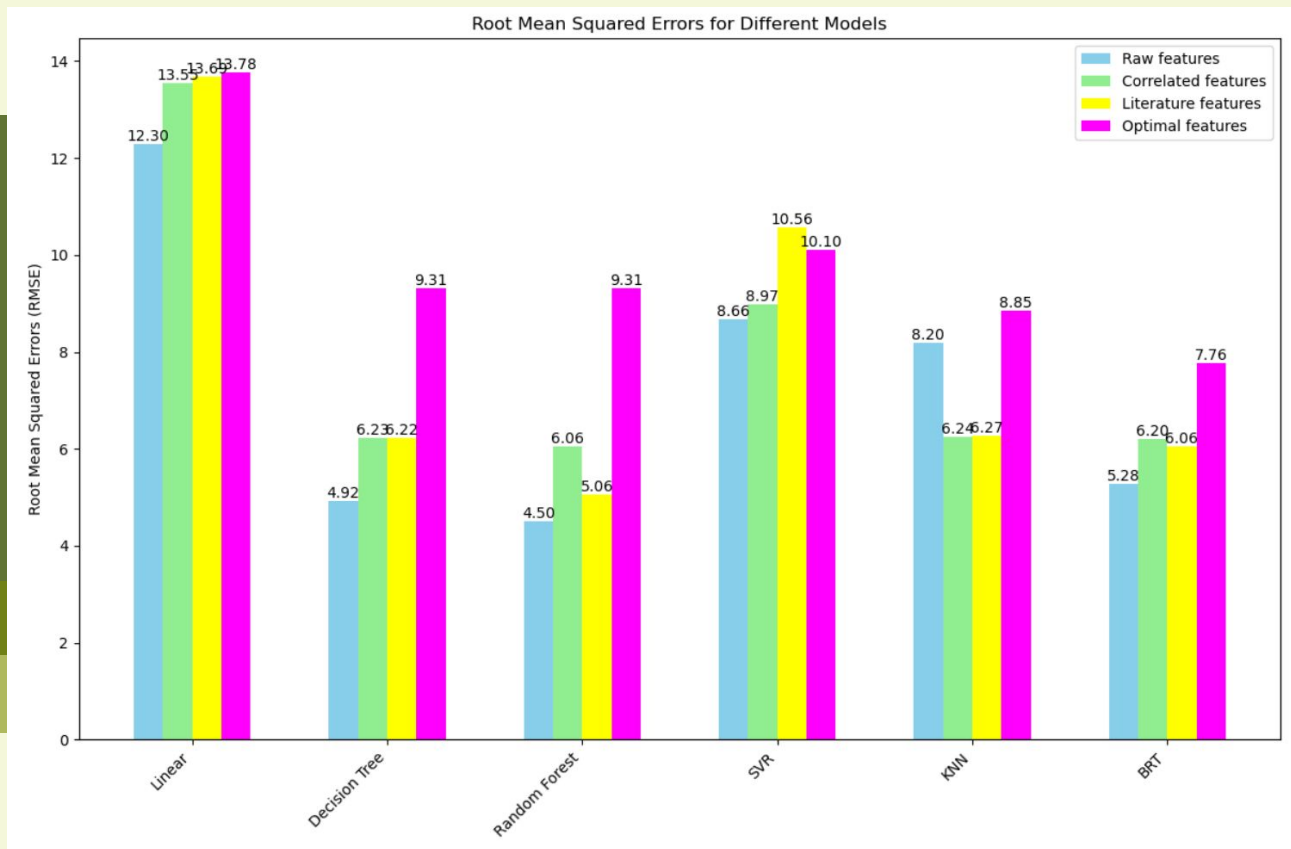
# Coefficient of Determination for Different Models and Feature Sets



# Coefficient of Determination for Different Models and Feature Sets Using 10-fold Cross Validation



# Root Mean Squared Error for Different Models and Feature Sets



# Root Mean Squared Error for Different Models and Feature Sets Using 10-fold Cross Validation

